

Weekly Self-Study Tasks to Complete

Phase 1 — Week 6: Introduction to Programming for Leaders

This document covers all of Week 6 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen Insurance Quote Page lab, where you will read the existing `app.js`, set a Chrome DevTools breakpoint, and adjust one rate in a lookup table. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

The week's single core video is deliberately concept-only: David Tucker's "JavaScript: The Big Picture" answers what JavaScript is, where it runs, and how new versions land, without asking you to write any code. The lab is where you'll touch real JavaScript. Every link and duration below has been verified against the source platform.

Topic	Material Description	Est. Time	Link / Reference
JavaScript Concepts	JavaScript: The Big Picture (David Tucker, Pluralsight, updated Nov 2, 2022) — watch the full course. Four modules: Why JavaScript? (~10 min), Where Is JavaScript Used? (~23 min), How Is JavaScript Versioned? (~17 min), plus a short intro and Next Steps. Concept-only — no coding exercises. Leaders-appropriate orientation to what the language is, where it runs, and how releases are governed.	~56 min	pluralsight.com — JavaScript: The Big Picture
Slide Skim	Review the week's slide deck at a high level. Confirm you recognize the	~5 min	See course material

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	terms variable, function, object, event listener, and breakpoint. No need to memorize syntax — this is a check that the instructor session and the Pluralsight course together left you fluent in the vocabulary you'll see in any JS file. Skim only.		

Practice Activities (1 hour)

Complete this week's hands-on lab: “Evergreen Insurance Quote Page — Rate Tweak.” The starter is the same Evergreen page you built up in Weeks 2–4. This week you will read the `app.js` file that powers the Calculate Premium button, pause it mid-execution using a Chrome DevTools breakpoint, step through the paused function, and then change one number in the pricing lookup table to see the UI update. The whole exercise takes about one hour.

In the lab you will:

- Open the Labfiles/evergreen-quote-rate-tweak folder in VS Code and launch the page with Live Server.
- Read `app.js` top-to-bottom and name each pattern: comments, `const` variables, event listeners, the `calculatePremium` function, the `ratePerThousand` object literal, the `||` fallback, DOM updates.
- Open Chrome DevTools → Sources tab, locate `app.js` in the file tree, and set a line breakpoint on `const rate = ratePerThousand[coverageType] || 1.0;`.
- Trigger the form (pick Auto, slider at \$50,000, click Calculate) and watch the page pause on your breakpoint.
- Hover live variables in the paused function — `coverageType`, `ratePerThousand`, `rate` — to see their current values in the tooltip.
- Use Step Over to advance line-by-line through `calculatePremium` and watch `rate` and `thousands` take on their values.
- In VS Code, change `ratePerThousand.auto` from `1.2` to `1.5`, save, and confirm the Auto \$50,000 premium moves from \$60 to \$75 on the page.

- Optional: change the “\$” label prefix to “Estimated: \$” in the DOM-update line and confirm the new text appears.

The lab instructions (with “Why this matters” callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for leaders who want to go deeper beyond the core curriculum. They sit outside the 4-hour weekly budget. None of these are required — you can absolutely stop at the Pluralsight course and the lab and have everything you need for the rest of Phase 1.

Topic	Material Description	Link / Reference
JavaScript (Hands-On)	Hands-On Introduction: JavaScript — Morten Rand-Hendriksen (LinkedIn Learning, Nov 9, 2022). 1h 43m Intermediate. A coding-forward introduction that uses the browser console as a learning environment. Recommended only if you want to try writing JavaScript yourself (Video Course)	linkedin.com/learning — Hands-On Introduction: JavaScript
JavaScript Reference	MDN JavaScript Guide — developer.mozilla.org. The canonical free reference for JavaScript syntax and semantics. Organized as an introduction, with deep-links to every keyword and built-in. Keep it open when you are reading a PR with unfamiliar syntax (Reference)	developer.mozilla.org — JavaScript Guide
JavaScript (Free Book)	Eloquent JavaScript, 4th edition — Marijn Haverbeke (2024). Free to read online under a Creative Commons license. Chapters 1–4 are the relevant read for this week: values and types, program	eloquentjavascript.net

Topic	Material Description	Link / Reference
	structure, functions, and data structures. The rest of the book goes much deeper and is optional (Book)	
Interactive	Introduction to JavaScript — Codecademy. Free interactive modules that run in the browser. Covers the same concepts as the LinkedIn Learning courses but with inline exercises you complete as you read. Useful if you learn by typing rather than watching (Interactive)	codecademy.com — Introduction to JavaScript
DevTools Reference	Pause your code with breakpoints — Chrome for Developers documentation. The official guide to every kind of breakpoint Chrome DevTools supports: line-of-code, conditional, logpoints, DOM-change, event-listener. The Sources panel you used in this week's lab has more features than we had time to cover (Reference)	developer.chrome.com — Breakpoints
Can I Use	Can I Use — up-to-the-day browser-support matrix for every JavaScript feature. Useful when an engineer tells you a feature "isn't safe to ship yet" — you can check the matrix for yourself (Tool)	caniuse.com